

EMA705 - Advanced Topics in Finite Elements

HW 2: Non-linear material models

Given: 8 April, 2026, due: **20 April., 2026.** Due on Canvas.

Problem (25 points)

Consider a 3D geometry in the reference configuration with dimensions $[0, 10] \times [0, 3] \times [0, 3]$, where the dimensions are in *cm* and the intervals lie on the lines defined by the orthonormal triad of vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

Using the framework of **finite-strain** mechanics, solve the following boundary value problem using the nearly-incompressible hyperelastic material models listed below:

Model 1: Neo-Hookean (commonly used for rubber-like materials, soft polymers, and foams at moderate strains):

Strain energy density function:

$$W = C_{10} (\bar{I}_1 - 3) + \frac{1}{D_1} (J - 1)^2$$

where $C_{10} = 1 \times 10^{10}$ Pa, $D_1 = 1 \times 10^{-10}$ Pa⁻¹ (i.e., $\mu = 2C_{10} = 2 \times 10^{10}$ Pa, $\kappa = 2/D_1 = 2 \times 10^{10}$ Pa). Here $\bar{I}_1 = J^{-2/3} I_1$ is the first deviatoric strain invariant and $J = \det(\mathbf{F})$.

Model 2 (Optional): Mooney-Rivlin (commonly used for filled rubbers, elastomeric seals, and biological soft tissues at moderate-to-large strains):

Strain energy density function:

$$W = C_{10} (\bar{I}_1 - 3) + C_{01} (\bar{I}_2 - 3) + \frac{1}{D_1} (J - 1)^2$$

where $C_{10} = 8 \times 10^9$ Pa, $C_{01} = 2 \times 10^9$ Pa, $D_1 = 1 \times 10^{-10}$ Pa⁻¹ (i.e., $\mu = 2(C_{10} + C_{01}) = 2 \times 10^{10}$ Pa, $\kappa = 2/D_1 = 2 \times 10^{10}$ Pa). Here $\bar{I}_2 = J^{-4/3} I_2$ is the second deviatoric strain invariant.

For the Neo-Hookean model, solve the following boundary value problem:

- Dirichlet boundary conditions: $\mathbf{u} = 0$ on $x_1 = 0$, $u_1 = 5$ on $x_1 = 10$. (approx. 50% axial strain)

Assume zero traction Neumann boundary conditions on boundaries where no boundary conditions explicitly specified. Use a 20 X 6 X 6 element mesh, and Lagrange basis.

Submit one ZIP file containing both the source code and a single PDF containing the following plots on Canvas:

- Source code for the 3D FE program. The code should be well commented to clearly show the various steps of the FE solution procedure, and the boundary conditions and initial conditions.
- Displacement plot of $\mathbf{u}(x)$ over Ω (deformed configuration) obtained for the Neo-Hookean material model showing the deformed geometry and the displacement contours. Also plot the mesh, if possible.